

Final Exam - 2H00

Exercise 1 1. Let a random variable X following the Poisson distribution. Present a simulation algorithm for this variable. Generate 4 realizations of X (for $\lambda = 1$).

2. Let the random variable X have pdf

$$f(x) = \begin{cases} \frac{1}{4} & \text{if } 0 < x < 1 \\ x - \frac{3}{4} & \text{if } 1 \leq x \leq 2 \end{cases}.$$

Write an algorithm of simulation of this random variable, using the acceptance-rejection method and generate 3 observations.

Use random numbers $\sim \mathcal{U}[0, 1]$:

0.0159 0.395 0.8145 0.8397 0.3090 0.8100 0.1634 0.7948 0.4279 0.7879 0.3352 0.4137

Exercise 2 We want to estimate the integral

$$I = \int_0^{\frac{\pi}{2}} \sin(x) dx.$$

1. a) Propose an estimator $\widehat{I}_N$ of I by mean sampling.

b) Calculate the theoretical variance of $\widehat{I}_N$.

2. a) Propose an estimator $\widehat{M}_N$ of I using importance sampling method.

b) Calculate the theoretical variance of $\widehat{M}_N$.

3. a) Compare the two theoretical variances. Comment the obtained results

b) Give an approximate number of iterations required to achieve an accuracy of the previous estimator ± 0.01 .

Exercise 3 In a higher education institution, the specialty studies take three years ; at the end of each year, each student has a probability 0.7 of advancing to the next year, a probability 0.2 of repeating the year, and a probability 0.1 of being expelled. We note states E and D resp. expulsion and diploma obtained.

1. The student's cursus can be modeled as a Markov chain with five states, provide its transition matrix, and draw the associated graph for this chain.

2. Calculate the probability that a student obtains his diploma based on his current state in the cursus.

3. Calculate the average duration of studies.

4. Give a simulation algorithm of this discrete time Markov chain.

Exercise 4 In a service company, where users arrive according to a Poisson process with rate $\lambda > 0$, management can choose between 3 models:

1. Installation of a server whose duration follows an exponential law with parameter 2μ .
2. Installation of 2 separate, identical servers, each with a service duration following an exponential law of parameter μ . Any new user chooses one of the servers with probability 0.5, and there is no exchange of potential users between the servers.
3. Installing two identical, non-separated servers. As soon as one of the servers is free, if there are users waiting, the first to arrive is served by the free server.

Assume that $\frac{\lambda}{2\mu} < 1$. Calculate the average time spent in the system (for the three systems in question). Which of the 3 systems performs best?

Recall:

$$p_n = \frac{1}{s!} \frac{1}{s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n p_0, \text{ for } n \geq s \text{ and } p_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!} \left(\frac{\lambda}{\mu} \right)^n + \frac{\left(\frac{\lambda}{\mu} \right)^s}{s! \left(1 - \frac{\lambda}{s\mu} \right)} \right]^{-1}.$$